Bioresource Technology 196 (2015) 169–175

Contents lists available at ScienceDirect

Bioresource Technology

journal homepage: www.elsevier.com/locate/biortech

# Treatment of petrochemical wastewater by microaerobic hydrolysis and anoxic/oxic processes and analysis of bacterial diversity

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## HIGHLIGHTS

- Microaerobic hydrolysis–acidification (MHA) was used as a pretreatment.
- MHA–A/O process was developed to treat actual petrochemical wastewater.
- The total COD removal efficiency was 72–79% and MHA accounted for 33–42%.
- The predominant genera in MHA, anoxic and oxic reactors were determined.

## ARTICLE INFO

*Article history:*
Received 20 June 2015
Received in revised form 20 July 2015
Accepted 21 July 2015
Available online 29 July 2015

*Keywords:*
Microaerobic hydrolysis
Petrochemical wastewater
Microbial community
Wastewater treatment

## ABSTRACT

Microaerobic hydrolysis–acidification (MHA)–anoxic–oxic (A/O) processes were developed to treat actual petrochemical wastewater. The results showed that the overall COD removal efficiency was 72–79% at HRT = 20 h, and MHA accounted for 33–42% of COD removal, exhibiting good efficiency of acidogenic fermentation. Ammonium removal was more than 94%. The main pollutants in the influent were identified to be benzene, ketone, alcohols, amine, nitrile and phenols by GC–MS, and the majority of pollutants could be removed by MHA–A/O treatment. *Proteobacteria* was the most dominant bacteria in the system, accounting for more than 55% of the reads. The predominant genera in MHA, anoxic and oxic reactors were *Anaerolineaceae* and *Sulfuritalea*, *Lactococcus* and *Blastocatella*, and *Saprospiraceae* uncultured and *Nitrosomonadaceae*, respectively. This treatment system exhibited good performance in degrading the complex compounds in the petrochemical wastewater.

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## 1. Introduction

A large amount of petrochemical wastewater was generated in the petroleum refining industry and the manufacturing processes of numerous organic chemicals and raw materials, which has become one of the most serious issues. Usually wastewater released from the petrochemical industry contains hazardous chemicals such as petroleum hydrogen, aromatic compounds, phenolic substances and heavy metals, which are highly toxic and biologically recalcitrant (Wang, 2002; Liu et al., 2014).

The treatment processes used for petrochemical wastewater generally include pretreatment to improve the biodegradation and reduce toxicity, followed by biological treatment involving activated sludge process, anoxic–oxic (A/O) process, fluidized bed reactor, membrane bioreactor and biofilm process (Wang et al., 2000; Guo et al., 2009). Among the biological treatment processes available, the A/O process is highly effective in removing organics and nutrients with low operational costs, and has been a promising alternative to conventional activated sludge process (Wang et al., 2002, 2004). The pretreatment processes include several physicochemical methods such as ultrasonic, flocculation, Fenton and ozone oxidation as well as anaerobic hydrolysis–acidification (Lin et al., 2001).

Anaerobic hydrolysis–acidification is the first two stages of anaerobic digester, serving for solubilization of complex macromolecular and particular organic compounds into micromolecular and simple soluble compounds such as volatile fatty acids (VFAs). Recent studies found that the hydrolysis–acidification process could occur under microaerobic conditions by adding a small amount of oxygen (air) into the reactor, which is termed as microaerobic hydrolysis–acidification (MHA) (Lim et al., 2014).

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http://dx.doi.org/10.1016/j.biortech.2015.07.087
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Studies showed that limited oxygen supply to anaerobic digester did not affect the methane production and organic substances removal (Diaz et al., 2011a). Moreover, Jenicek et al. (2008) found that the application of microaerobic condition was efficient to increase the biodegradation of solid waste. The studies performed by Lim and Wang (2013) showed that microaerobic condition could increase the solubilization and acidification efficiencies, and the conversion rate of other short chain fatty acids to acetate in the anaerobic co-digester of brown water and food waste. This might be due to the enhanced metabolic activities of facultative hydrolytic and acidogenic bacteria under microaerobic conditions. Another reported benefit of microaeration was to accelerate the production of exoenzymes that conducted the degradation of slowly biodegradable organics (Lim and Wang, 2013).

Moreover, it was reported that sulfide oxidizing bacteria, which are responsible for oxidizing sulfide to sulfite with oxygen as electron acceptor, exhibited higher activities in microaerobic conditions than that in anaerobic digester (Jenicek et al., 2011). As a result, the production of hydrogen sulfide ($H_2S$) content could be decreased effectively. $H_2S$ is a toxic gas with rotten egg odors, is highly soluble and causes corrosion of the equipment. Literature demonstrated that the concentration of $H_2S$ in the biogas was reduced by more than 98% under microaerobic conditions (Diaz et al., 2011b). Up to now, most of studies related microaeration focused on the digester of solid waste, there is a little of information about the application of MHA process on treating industrial wastewater, and even less on petrochemical wastewater. Chen et al. (2011) studied the treatment of pharmaceutical wastewater using combined anaerobic/microaerobic and two-stage aerobic processes. MHA was used to enhance the biodegradability of pharmaceutical wastewater and benefit the subsequent aerobic treatment.

In this study, MHA and A/O processes were developed to treat the real petrochemical wastewater from a large petrochemical company in northeast China. Experiments were conducted to: (1) evaluate the performance of the system in removing organics, nitrogen and phosphorus compounds; (2) analyze the composition of organics in influent and effluent; (3) identify the bacterial community of the sludge in each tank. Knowledge of the microbial community structure is helpful for a better control of the biological processes. With this information it was aimed to deepen our understanding about MHA and provide a new choice for industrial wastewater treatment.

## 2. Methods

### 2.1. Experimental set-up

The experimental set-up was built in an Integrated Petrochemical Wastewater Treatment Plant (IPWWTP) in northeast China. It consisted of a MHA reactor, an A/O reactor and a settling tank, with an effective volume of 33 L, 43 L and 28 L, respectively (Fig. 1). The MHA reactor was divided into four compartments. Air was supplied into the bottom of the first three compartments through aquarium-type diffuser with a flow rate of 0.3 L/h. The sludge from the fourth chamber was recycled into the first chamber. The A/O reactor consisted of an anoxic and two oxic compartments, with a volume ratio of 1:2:2. Dissolved oxygen (DO) concentration in the oxic tanks was maintained at 4–6 mg/L by adjusting the input aeration flowrate (Wang and Yang, 2004). An electric agitator was equipped in the anoxic tank to provide a mixture of sludge and wastewater. Sludge from the settling tank was recycled into the anoxic tank with a ratio of 100%.

### 2.2. Wastewater and characteristics

The wastewater used in this study was the influent of IPWWTP, which consists of petrochemical industry wastewater and domestic wastewater with a ratio of 3:1. The industrial wastewater came from more than 70 sources involving fertilizer plant, oil refinery plant, resin factory, acrylonitrile butadiene styrene (ABS) plants, etc. It was pre-treated in the plants and effluent was discharged into the IPWWTP. The domestic wastewater is the used water from houses, commercial and schools in the area around IPWWTP.

The experiment was divided into two working periods. The reactor was operated by feeding the influent of IPWWTP in the first period and industrial wastewater alone in the second period. Table 1 presents the characteristics of wastewater. The concentrations of COD and sulfate ($SO_4^{2-}$) in the influent fluctuated significantly. The volatile phenolic compounds (VPCs) and benzene compounds (BCs) were mainly from the industrial wastewater. There is not great difference in total nitrogen (TN) and total phosphorus (TP) concentration for the wastewater at two phases.

### 2.3. Start-up and operation of MHA–A/O reactor

The MHA–A/O reactor was started up by seeding the recycled sludge taken from the recirculation line of the IPWWTP at an initial hydraulic retention time (HRT) of 48 h. The HRT was decreased gradually by using the COD removal (15–30% for MHA and 65–70% for A/O reactors) as a reference parameter until stable operational conditions were achieved. After a month, the system was started-up successfully and operated for nearly a half of year, from Aug 2014 to Jan 2015, at HRT of 20 h (12 h and 8 h in the MHA and A/O reactor, respectively). The sludge retention time was set at 30 d. The loading rate of the MHA and A/O reactor ranged 0.2–0.7 and 0.1–0.4 g COD/L d, respectively. The temperature of the system ranged 22–28 °C.

During steady-state operation, the performance of the system was evaluated by measuring the following parameters in influent and effluent periodically: COD, TN, ammonium ($NH_4$-N), nitrate ($NO_3$-N), nitrite ($NO_2$-N), TP, $UV_{254}$, VFA, $SO_4^{2-}$ and sulfide ($S^{2-}$). All the samples were settled and filtered using 0.45 μm syringe filters before analysis. The distribution of COD and $NH_4$-N at each tank of the reactors, $BOD_5$, total suspended solids (TSS) and volatile

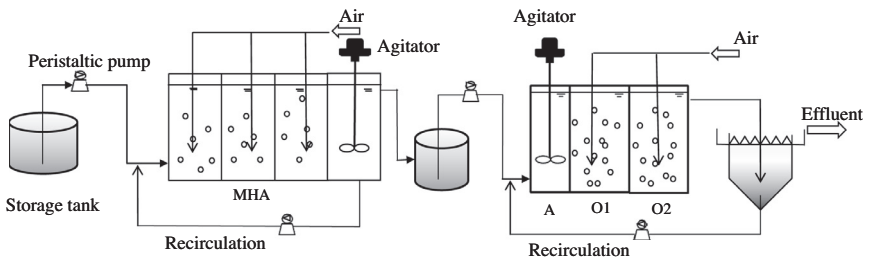



**Fig. 1.** Schematic diagram of the experimental set-up.

**Table 1**
Characteristics of the influent.

| Phase | COD (mg/L) | $NH_4^+$-N (mg/L) | TP (mg/L) | TN (mg/L) | $UV_{254}$ | $SO_4^{2-}$ (mg/L) | VPC (mg/L) | BCs (mg/L) |
|---|---|---|---|---|---|---|---|---|
| I | 348 ± 59 | 25.8 ± 5.8 | 1.3 ± 0.6 | 30.5 ± 9.3 | 1.4 ± 0.3 | 380 ± 115 | 13.0 ± 2.5 | 16.9 ± 4.7 |
| II | 529 ± 30 | 28.5 ± 5.4 | 1.4 ± 0.6 | 35.2 ± 6.2 | 2.2 ± 0.3 | 552 ± 162 | 17.5 ± 3.6 | 22.5 ± 6.9 |

VPCs: volatile phenolic compounds; BCs: benzene compounds.

suspended solids (VSS) concentrations were detected 2–3 times every month. On the day 100, the sludge sampled from the MHA, anoxic and oxic tanks were taken for microbial community analysis, effluent was identified.

### 2.4. Analytical methods

The composition of organic pollutants of the influent and concentrations of COD, $BOD_5$, $NH_4$-N, $NO_3$-N, $NO_2$-N, TN and TP were analyzed following the Standard Methods. The contents of $SO_4^{2-}$, $S^{2-}$, VFA, VPCs and BCs were determined using barium chromate spectrophotometry, methylene blue method, titration method, 4-AAP spectrophotometric method and gas chromatography, respectively. DO and Oxidization Reduction Potential (ORP) were measured by a portable DO meter (WTW 340i). pH was measured by a SX721 pH meter. TSS and VSS were detected using gravimetric method. The dissolved organic matter in effluent was characterized by three dimensional excitation-emission matrix (3D-EEM) fluorescence spectroscopy (Hitachi F-7000) as described in previous studies (Chu and Wang, 2011).

The composition of organic matters in the wastewater was identified by GC/MS analyzer (Agilent 7890A/5975C). Solid phase extraction was used to extract the organic substances from wastewater samples (Reeko automatic solid-phase extraction apparatus). Before use, the C18E cartridges (Sepax Generik) were conditioned by passing methanol and deionized water. The acidified samples were loaded into the cartridges and the cartridges were dried with a stream of nitrogen of 99.999% purity. Then dichloromethane were loaded to extract the organic compounds retaining in the cartridges. Finally, the extracts were evaporated and reconstituted with dichloromethane to a volume of 1 mL prior to analysis.

The stationary phase of the GC capillary column (50 m × 0.25 mm × 0.25 m) was OV-101. The analyses were carried out in a split ratio of 1:25 and inlet temperature of 280 °C. Helium was used as carrier gas with a flow rate of 0.6 mL/min. The applied temperature program was followed by retaining at 55 °C for 4 min and then increasing to 280 °C with an increment of 10 °C/min. The MS detector was operated in electron ionization (70 eV) using a scan mode of 29–350 *m*/*z*. The temperature of the MS ion source was 200 °C.

The Illumina MiSeq sequencing (Schmidt et al., 2013) was applied to analyze the microbial consortium in MHA, anoxic and oxic tank, respectively. The procedures are as follows: (1) DNA of sludge samples was extracted using E.Z.N.A. soil DNA extraction kit (OMEGA) and detected using 1% agarose gel electrophoresis; (2) PCR amplification was performed using specifically synthesized primers with barcode. Amplicons were purified using an AxyPrep™ DNA Gel Extraction Kit (AXYGEN, USA) and eluted with Tris HCl. Replicate amplicons were pooled and visualized on 2.0% agarose gels using SYBR Safe DNA gel stain in 0.5× TBE; (3) amplicons were quantified by QuantiFluor™-ST blue fluorescence quantitative system (Promega) and homogenized. Then the amplicon sequencing was performed using an Illumina MiSeq PE250 system by Shanghai Majorbio Bio-pharm Biotechnology (Shanghai, China); (4) results were optimized and Operational Taxonomic Units (OTU) clustering and taxonomy analysis was carried out at a 97% similarity level.

## 3. Results and discussion

### 3.1. Performance of MHA–A/O reactor

Under steady-state conditions, the average pH value in the influent and effluent of MHA and A/O reactor was 7.96, 7.71 and 7.48, respectively. The sludge concentration in MHA and A/O reactor ranged 5300–6500 mg TSS/L and 4400–5000 mg TSS/L, with VSS/TSS ratio of around 42.7% and 46.7%, respectively. The ORP was applied to monitor the level of oxidant in MHA tank because it could sensitively reflect the changes of DO at low levels. The ORP of MHA reactor varied between −180 and −300 mV, which favors the acid fermentation.

Fig. 2 shows the performance of the system in terms of COD and ammonium concentrations of influent and effluent and removal efficiencies. Although the COD concentration of influent fluctuated,

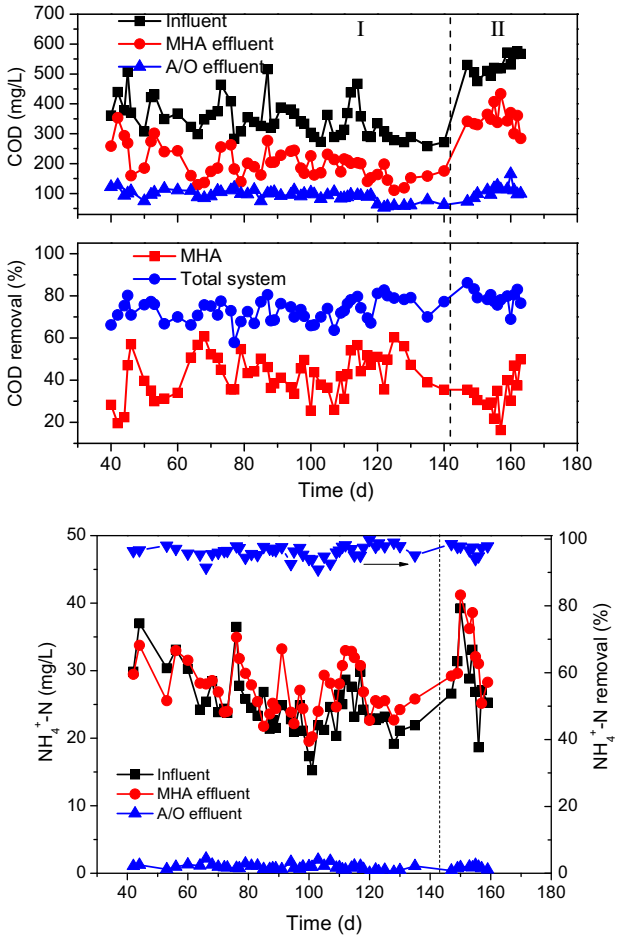



**Fig. 2.** Time course of COD and $NH_4^+$-N concentration of the influent and effluent and their removal efficiencies.

the effluent COD remained stable, indicating the system has good ability to withstand fluctuation in influent substrate concentration. The effluent COD reached 97 ± 19 mg/L and 109 ± 22 mg/L at phase I and II, resulting in COD removal efficiency of 72 ± 6% and 79 ± 4%, respectively. Among them, MHA reactor contributed 42–33% COD removal. The overall COD removal at phase II was even slightly higher than that at phase I.

The ammonium removal was higher than COD removal. The effluent $NH_4^+$-N concentration maintained at low levels (1.2 ± 1.0 mg/L), with removal efficiency of 94.6 ± 0.6%, indicating the activity of ammonium oxidation bacteria in A/O reactor was high during the overall operational period, even at second phase feeding petrochemical wastewater alone. The increased $NH_4$-N concentration in the MHA effluent might be due to ammonification of organic nitrogen compounds, which were identified by the following GC/MS analysis. The N compounds in the effluent mainly existed in the form of nitrate (13.6 ± 4.5 mg/L) and less than 0.05 mg/L of nitrite was detected, indicating nitrification occurs completely. The concentrations of effluent TN and TP were 17.5 ± 5.2 mg/L and 0.8 ± 0.4 mg/L, respectively with removal efficiencies of 55% and 48% on average. $UV_{254}$ of effluent decreased to around 0.5 at phase I and 0.8 at phase II, with total removal efficiency of around 54%, which indicated that the content of aromatic compounds and double bond structure compounds in the influent was reduced significantly.

Fig. 3 shows the distribution of COD and $NH_4$-N at each tank of the system. Both MHA and A/O reactor contributed to COD degradation, while $NH_4$-N was mainly removed in the oxic tank by aerobic nitrification. The reduced $NH_4$-N concentration in anoxic tank was due to 100% recycling of sludge mixture.

In this study, MHA reactor was employed as pretreatment to improve the biodegradability of petrochemical wastewater and inhibit the production of toxic gas $H_2S$. Results showed that the value of $BOD_5$/COD ratio increased from 0.27 ± 0.18 of influent to 0.34 ± 0.14 of MHA effluent, and decreased significantly to less than 0.03 of A/O effluent. This indicated that MHA could improve the biodegradability of petrochemical wastewater and almost all of the biodegradable compounds were removed by A/O treatment. The detected VFA content in the MHA effluent was 2.33 ± 0.98 mmol/L (Fig. 4). The ratio of VFA to COD increased from 0.37–0.57 of the influent to 1.02–1.07 of the MHA effluent, indicating the good efficiency of acidogenic fermentation in MHA reactor. During microaerobic operation, the oxygen supply should be controlled carefully since the intensity of microaeration determines the hydrolysis rates and acidification degree. Zhu et al. (2009) reported a positive effect on hydrolysis efficiency but a negative effect on acidogenesis when a microaeration intensity of 74 L air/TSS d was used to anaerobic digester of vegetables and flower wastes. Johansen and Bakke (2006) found a 55–60% increase in

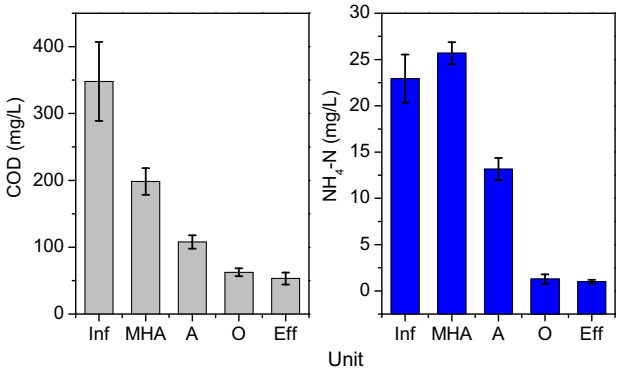



**Fig. 3.** Changes in COD and concentrations at different parts of reactor.

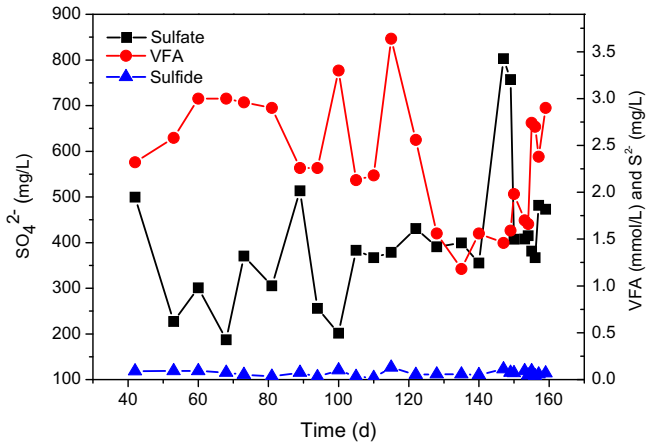



**Fig. 4.** Variation of effluent sulfate, sulfide and VFA with time.

hydrolysis of primary sludge at a microaeration intensity of 33.3 L air/TSS d. In this study, the activities of hydrolytic and acidogenic bacteria remained high when microaeration intensity of around 33.6 L air/TSS d (5.5 mL $O_2$/L d) was applied.

$SO_4^{2-}$ and $S^{2-}$ concentrations in the MHA effluent were around 400 mg/L and 0.065 mg/L, respectively (Fig. 4). The low level of $S^{2-}$ was attributed to the oxidation of $S^{2-}$ to S by sulfide oxidizing bacteria, a genus of which was identified and proved dominant in the MHA reactor in the following sequencing analysis. As a result, the amount of odors gas $H_2S$ produced could be decreased, which is one of the benefits of MHA compared to anaerobic digester.

### 3.2. Characterization of organic pollutants in influent and effluent

From GC–MS analysis as shown in Fig. 5, the influent consisted of many kinds of organic pollutants such as benzene, ketone, ethanol, esters, amides, nitriles and phenols, etc. The variety and abundance of organic pollutants decreased significantly by MHA–A/O treatment. According to 3D-EEM spectrum (Fig. 6), the dominant peak A related to tyrosine and peak B related to soluble microbial products (SMP) were identified in the A/O effluent. This indicated that the effluent mainly consisted of protein-like substance with a small SMP-like substance. The results were comparable to other

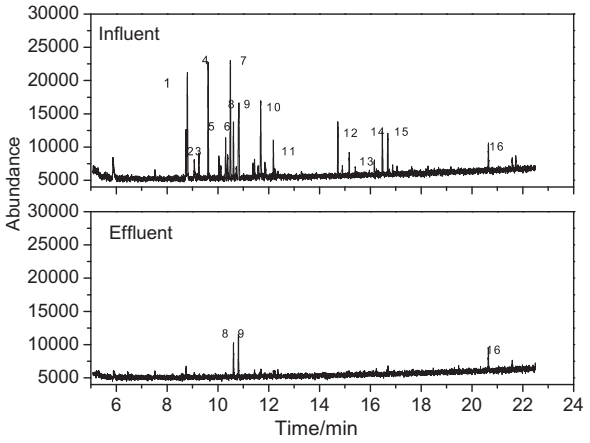



**Fig. 5.** GC–MS analysis of influent and effluent (1: phenol, 2: butanedinitrile, 3: carbethoxyhydrazine, 4: 2-ethylhexyl alcohol, 5: acetophenone, 6: ethylene chlorohydrin, 7: 2-(1-cyclohexenyl) ethylamine, 8: 2-isopropyl benzene, 9: 1,4-cyclohexane two methylamine, 10: 3-methyl-1-cyclohexene, 11: methyl vinyl nitrosamine, 12: 2-bromide ethanol, 13: methyl phenethylamine, 14: N-(2-phenyl) acetamide, 15: phenylethylamine, 16: 3-nitrostyrene).

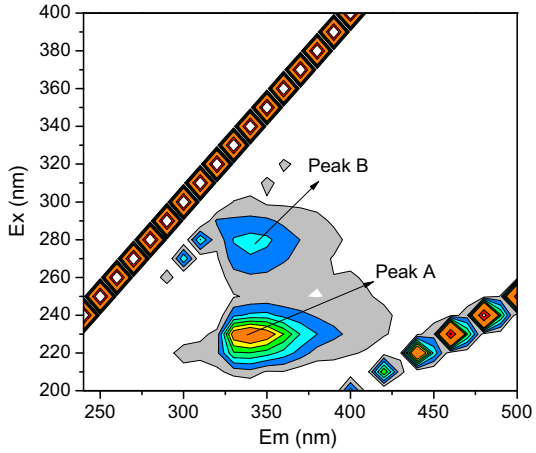



**Fig. 6.** Fluorescence 3D-EEM spectrum of A/O effluent (diluted by 20 times).

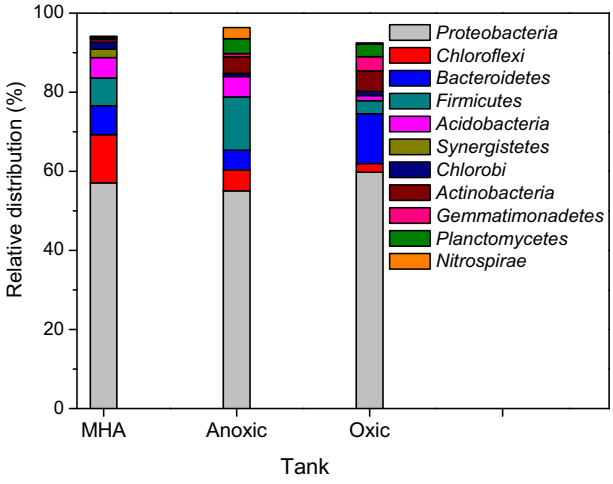



**Fig. 7.** Relative abundance of the major bacterial community at phylum levels (only higher than 1% was present).

studies. Llop et al. (2009) have used a MBR to treat petrochemical wastewater from various sources such as refinery process, the olefin process, etc. A COD reduction of 84% was achieved at HRT of 12.5 h and 5 h. GC–MS analysis demonstrated that most of the identified organic compounds, involving hydrocarbons, benzene, phenol and ester derivates, were removed partially and completely (80–100%) by MBR treatment. Estrada-Arriaga et al. (2012) employed the anaerobic and aerobic fluidized bed reactors to treat petrochemical wastewater. The overall COD removal was over 96% and all aromatic compounds were removed at a total HRT of 4.9 h.

### 3.3. Bacterial community in MHA, anoxic and oxic tanks

The Illumina MiSeq pyrosequencing analysis yielded 41,772 sequences from three genomic DNA samples after removal of short and low-quality reads. The numbers of OTUs (3% dissimilarity) for each library are shown in Table 2. The high Good's coverage of the samples demonstrated that the results of sequencing analysis could reflect the real profile of microorganisms. The Chao1 and Ace estimators indicate the community richness, while Shannon index and Simpson indices represent the community diversity. The higher the Shannon index and the lower the Simpson index, the more community diversity exists. The data clearly showed that anoxic sludge exhibited high community richness and diversity, followed by oxic and microaerobic sludge.

Fig. 7 shows the relative abundance of the major bacterial community at phylum levels. *Proteobacteria* is the most dominant in all sludge samples, accounting for more than 55% of the reads. The Gram-negative *Proteobacteria* are the major group of bacteria, involving a wide variety of aerobic, anaerobic and facultative bacteria. They are able to degrade a broad spectrum of organic contaminants and enable biological nitrogen and phosphorous removal, which are specially suited for wastewater treatment (Snaidr et al., 1997). *Chloroflexi*, *Bacteroidetes* and *Firmicutes* were highly enriched in the MHA and anoxic tanks. Those microorganisms were commonly observed in the anaerobic digester reactor. Studies by Tang et al. (2004) revealed that bacteria belonging to phylum *Firmicutes* were dominant under anaerobic and microaerobic conditions for treatment of municipal solid waste. Lim et al. (2014) found that microaeration conditions led to a more diverse bacterial community for treating brown water and food waste. Bacteria affiliated with phyla *Firmicutes* and *Bacteroidetes* represent the dominant phylogenic group. It should be noted that the major role the anaerobes *Bacteroidetes* in the fermentation system is to break down macromolecules such as protein, starch, cellulose and fiber. Some species affiliated with phylum *Firmicutes* are known to produce extracellular enzymes such as cellulose, lipase and protease. This is important for hydrolyzation and utilization of the refractory chemicals in petrochemical wastewater.

The relative abundance of the dominant genera identified in the system is shown in Table 3. The predominant genera in MHA, anoxic and oxic tanks were *Anaerolineaceae* and *Sulfuritalea*, *Lactococcus* and *Blastocatella*, and *Saprospiraceae* uncultured and *Nitrosomonadaceae* uncultured, respectively. Bacteria belonging to *Xanthomonadales* and *Defluviicoccus* were highly enriched in the total system. The bacterial composition of the sludge depends on the influent characteristics and operation conditions. Shokrollahzadeh et al. (2008) have investigated the bacterial

**Table 2**
Characteristics of community diversity in MHA, anoxic and oxic tanks.

| Sample | Reads | OTU | Coverage | Ace | Chao1 | Shannon | Simpson |
|---|---|---|---|---|---|---|---|
| MHA | 11,322 | 375 | 0.9932 | 434 | 443 | 4.2 | 0.0408 |
| Anoxic | 18,016 | 750 | 0.9927 | 840 | 853 | 5.32 | 0.0134 |
| Oxic | 12,434 | 627 | 0.9883 | 738 | 768 | 4.65 | 0.0371 |

**Table 3**
The dominant genera identified in the system, phylogenetic affiliation and abundance.

| Organisms | Phylum | Abundance (%) | | |
|---|---|---|---|---|
| | | MHA | Anoxic | Oxic |
| *Anaerolineaceae* uncultured | *Chloroflexi* | 7.10 | 2.12 | 0.60 |
| *Sulfuritalea* | *Proteobacteria* | 5.60 | 0.93 | 0.57 |
| *Ottowia* | *Proteobacteria* | 4.16 | 0.75 | 0.36 |
| *Blastocatella* | *Acidobacteria* | 3.46 | 3.01 | 0.47 |
| *Thermomonas* | *Proteobacteria* | 2.34 | 0.08 | 0.06 |
| *Xanthomonadales* uncultured | *Proteobacteria* | 1.40 | 1.86 | 2.75 |
| *Defluviicoccus* | *Proteobacteria* | 1.06 | 1.33 | 4.20 |
| *Nitrosomonadaceae* uncultured | *Proteobacteria* | 1.04 | 2.86 | 5.02 |
| *Novosphingobium* | *Proteobacteria* | 0.65 | 0.76 | 0.42 |
| *Bradyrhizobium* | *Proteobacteria* | 0.44 | 1.74 | 0.30 |
| *Lactococcus* | *Firmicutes* | 0.37 | 7.64 | 1.09 |
| *Hyphomicrobium* | *Proteobacteria* | 0.35 | 1.29 | 0.61 |
| *Xanthobacter* | *Proteobacteria* | 0.28 | 0.71 | 1.50 |
| *Erythrobacteraceae* unclassified | *Proteobacteria* | 0.23 | 2.44 | 0.92 |
| *Bacillus* | *Firmicutes* | 0.17 | 2.22 | 0.26 |
| *Saprospiraceae* uncultured | *Bacteroidetes* | 0.17 | 1.80 | 9.84 |
| *Pseudomonas* | *Proteobacteria* | 0.16 | 0.77 | 0.14 |
| *Nitrospira* | *Nitrospirae* | 0.15 | 2.83 | 0.37 |
| *Thiobacillus* | *Proteobacteria* | 0.05 | 1.59 | 0.10 |
| *Nitrosomonas* | *Proteobacteria* | 0.01 | 0.98 | 0.18 |

diversity of sludge in a petrochemical wastewater treatment plant using activated sludge process. The contaminants in influent were mainly composed of alkanes, aromatic and polycyclic hydrocarbons. The culturable bacterial population isolated from the activated sludge mainly belonged to genera *Pseudomonas*, *Flavobacterium*, *Comamonas*, *Cytophaga*, *Sphingomonas*, *Acidovorax* and *Bacillus*.

Members of the genus *Anaerolineae* are known as semi-syntrophic and fatty acids-oxidizing bacteria, degrading carbohydrate cooperatively with hydrogenotrophic methanogens (Narihiro et al., 2012). Some strains of genus *Sulfuritalea* are the obligate and facultative sulfur chemolithoautotrophs, which could oxidize inorganic sulfur compounds (Watanabe et al., 2014). This might be the reason why the low sulfide ions concentration was detected in MHA effluent.

*Lactococcus* is a genus of lactic acid bacteria that were known as homofermentors, producing a single product, lactic acid as the major only product of glucose fermentation. Some species of genus *Blastocatella* are aerobic, chemoorganotrophic bacterium with strictly respiratory type of metabolism (Foesel et al., 2013). Xia et al. reported that members of genus *Saprospiraceae* were detected in sludge flocs in an activated sludge treatment plant which plays an important role in protein degradation by the production of extracellular enzymes (Xia et al., 2008). Some species of the genus *Defluviicoccus* were related to glycogen accumulating organisms that are able to anaerobic uptake of acetate and propionate as carbon source (Dai et al., 2007). The bacteria of genus *Xanthomonadales* belonged to the heterotrophic denitrifiers, which were detected in a constructed wetland and played an important role in denitrification (Chon et al., 2010).

Among the detected genera (Table 3), a strain of genus *Ottowia* enriched in MHA tank was isolated from a phenol-degrading colony enriched from coking wastewater activated sludge (Cao et al., 2014). *Xanthobacter* strains enriched in oxic tank have been reported to degrade chlorinated hydrocarbon (Ensign et al., 1992) and thiophene (Padden et al., 1997) compounds, some species of genus *Bacillus* enriched in anoxic tank could degrade diesel oil (Azmatunnisa et al., 2015) and *Novosphingobium* strains could degrade aromatic compounds (Balkwill et al., 1997). With the combining effects of varied bacteria in different tanks, the system was able to degrade the complex hazardous chemicals in the petrochemical wastewater.

In terms of nitrifiers and denitrifiers, the ammonium-oxidizing bacteria *Nitrosomonadaceae* uncultured and *Nitrosomonas*, and nitrite-oxidizing bacteria *Nitrospira* were mainly detected in the A/O reactor. Denitrifiers *Hyphomicrobium*, *Pseudomonas*, *Thiobacillus* and *Bacillus* were highly enriched in the anoxic tank. This favors the biological nitrogen removal in the A/O reactor by nitrification and denitrification.

Based on the studies, the treatment processes of IPWWTP were modified. The petrochemical wastewater and domestic wastewater was treated separately. MHA was used as pretreatment for petrochemical wastewater instead of anaerobic hydrolysis–acidification. That is, the petrochemical wastewater with a capacity of 60,000–80,000 $m^3$/d was treated by MHA–A/O processes followed by ozonation polishing, while A/O processes were applied for domestic wastewater treatment (20,000–30,000 $m^3$/d). The IPWWTP is being reconstructed and its performance will be evaluated in near future.

## 4. Conclusions

The combined processes of MHA pretreatment with A/O reactor were used to treat petrochemical wastewater. The following results were drawn: (1) although influent COD fluctuated, effluent COD remained stable. The total COD removal efficiency was 72–79% and contribution from MHA was 33–42%. (2) The identified organic pollutants in the influent include benzene, ketones, alcohols, esters, amides, nitriles and phenols, etc. The variety and abundance of organic pollutants reduced significantly by MHA–A/O treatment. (3) The predominant genera in MHA, anoxic and oxic reactors were *Anaerolineaceae* and *Sulfuritalea*, *Lactococcus* and *Blastocatella*, and *Saprospiraceae* uncultured and *Nitrosomonadaceae*, respectively.

## Acknowledgements

The research was supported by the Program for Changjiang Scholars and Innovative Research Team in University (IRT-13026). The authors would also like to thank the financial support provided by National S&T Major Project (Grant No. 2012ZX07201-005-06-01) and the National Natural Science Foundation of China (51178241; 51338005).

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